

Technical Documentation: Artificial Stress, Damping, and Pressure Terms in the Meshfree Cutting Solver

1 Introduction

This solver is a 2D, particle-based, Updated Lagrangian meshfree method (SPH-like) for large-deformation cutting. The three documented components are:

- **Pressure p :** computed per particle from a linear elastic equation of state (EOS) and coupled into the stress divergence as an isotropic term.
- **Artificial stresses $\mathbf{R}$:** a Monaghan-style tensile-instability correction, added inside the stress divergence operator to introduce repulsion under tension.
- **Damping / dissipation:** primarily (i) Monaghan artificial viscosity (compression-activated numerical damping) added directly to acceleration, and (ii) penalty contact + Coulomb friction forces from the tool, added as external forces. A further stabilizer, XSPH, smooths advection velocities.

The time integration is explicit and second-order (leapfrog predictor/corrector) where all right-hand-side (RHS) terms are assembled at the half step.

2 Methodology

2.1 Spatial discretization

The domain is discretized by particles indexed by $i = 1, \dots, N$ with state variables

$$(x_i, y_i, \rho_i, h_i, \mathbf{v}_i, \mathbf{S}_i, T_i), \quad \mathbf{v}_i = (v_{x,i}, v_{y,i}). \quad (1)$$

Here ρ_i is density, h_i is smoothing length, $\mathbf{S}_i$ is deviatoric stress (components $S_{xx}, S_{xy}, S_{yy}, S_{zz}$), and T_i is temperature.

2.1.1 Neighbor sets (Verlet lists)

Each particle i maintains a neighbor list $\mathcal{N}(i)$ built by spatial hashing into a uniform grid each step. Neighbors are included when

$$\|\mathbf{x}_i - \mathbf{x}_j\|^2 \leq (2h_i)^2. \quad (2)$$

2.1.2 Kernel and gradients

A 2D cubic spline kernel $W_{ij} = W(\mathbf{x}_i - \mathbf{x}_j, h_i)$ and its gradient ∇W_{ij} are used (support radius $2h$). Define

$$\mathbf{x}_{ij} = \mathbf{x}_i - \mathbf{x}_j, \quad r_{ij} = \|\mathbf{x}_{ij}\|, \quad q = \frac{r_{ij}}{h_i}. \quad (3)$$

Then W_{ij} and $\nabla W_{ij} = (\partial_x W_{ij}, \partial_y W_{ij})$ are evaluated by the cubic spline formulas.

2.1.3 CSPM gradient correction (optional)

An optional corrective SPH procedure (CSPM) replaces ∇W_{ij} with a corrected gradient to improve consistency. Define particle “volume” $V_j = m_j/\rho_j$ and moment matrix

$$\mathbf{B}_i = \sum_{j \in \mathcal{N}(i)} (\mathbf{x}_j - \mathbf{x}_i) \otimes \nabla W_{ij} V_j. \quad (4)$$

Then

$$\nabla W_{ij}^{corr} = \nabla W_{ij} \mathbf{B}_i^{-1}. \quad (5)$$

2.2 Temporal discretization (leapfrog predictor/corrector)

Let u denote any particle state component (e.g., x , ρ , v_x , S_{xx}). With timestep Δt , the scheme is:

Predict to half step

$$u_i^{n+\frac{1}{2}} = u_i^n + \frac{\Delta t}{2} \dot{u}_i^n. \quad (6)$$

Compute RHS at half step All $\dot{u}^{n+\frac{1}{2}}$ terms are assembled using the half-step state.

Correct to full step

$$u_i^{n+1} = u_i^n + \Delta t \dot{u}_i^{n+\frac{1}{2}}. \quad (7)$$

2.3 Per-step coupling procedure

At each time step, the solver performs the following operator-split procedure:

1. Build $\mathcal{N}(i)$ and precompute $W_{ij}, \nabla W_{ij}$ (SPH or CSPM).
2. Predictor step (6).
3. Reset RHS accumulators $\dot{u}_i \leftarrow 0$.
4. Apply contact + friction forces from tool (external forces).
5. Assemble continuum RHS:
 - (a) EOS pressure p .
 - (b) Artificial stress $\mathbf{R}$.
 - (c) Stress divergence (includes p and $\mathbf{R}$).
 - (d) Velocity gradient.
 - (e) Artificial viscosity (adds to acceleration).
 - (f) XSPH (adds to kinematic advection).
 - (g) Jaumann deviatoric stress rate.
 - (h) Continuity $\dot{\rho}$.
 - (i) Momentum $\dot{\mathbf{v}}$.
 - (j) Advection $\dot{\mathbf{x}}$.
6. Corrector step (7).
7. Plasticity update and boundary-condition enforcement.

3 Derivations

3.1 Pressure term

3.1.1 EOS formulation

Pressure is computed per particle by a linear elastic EOS with bulk modulus $K(T)$ and reference density $\rho_0(T)$:

$$c_0(T) = \sqrt{\frac{K(T)}{\rho_0(T)}}. \quad (8)$$

$$p = c_0(T)^2 (\rho - \rho_0(T)). \quad (9)$$

Definitions

- p : pressure (scalar).
- $K(T)$: bulk modulus as a function of temperature.
- $\rho_0(T)$: reference density as a function of temperature.
- $c_0(T)$: reference wave speed used in the EOS.

3.1.2 Coupling into stress used for divergence

The solver stores deviatoric stress $\mathbf{S}$ and forms in-plane “total” stress components as

$$\sigma_{xx} = S_{xx} - p, \quad \sigma_{yy} = S_{yy} - p, \quad \sigma_{xy} = S_{xy}. \quad (10)$$

3.2 Artificial stress (Monaghan tensile-instability correction)

3.2.1 Local rotation to a principal-like frame

Define stress components s_{xx}, s_{xy}, s_{yy} as in (10). A rotation angle is computed:

$$\theta = \frac{1}{2} \text{atan2}(2s_{xy}, s_{xx} - s_{yy} + \epsilon). \quad (11)$$

Let $c = \cos \theta$ and $s = \sin \theta$. The rotated normal stresses are

$$\tilde{s}_{xx} = c^2 s_{xx} + 2cs s_{xy} + s^2 s_{yy}, \quad (12)$$

$$\tilde{s}_{yy} = s^2 s_{xx} - 2cs s_{xy} + c^2 s_{yy}. \quad (13)$$

3.2.2 Tensile-only repulsive correction

Let $\epsilon_{AS} > 0$ be the artificial stress coefficient and ρ the density. The correction acts only in tension:

$$\tilde{R}_{xx} = \begin{cases} -\epsilon_{AS} \tilde{s}_{xx} \rho^{-2}, & \tilde{s}_{xx} > 0, \\ 0, & \tilde{s}_{xx} \leq 0, \end{cases} \quad (14)$$

$$\tilde{R}_{yy} = \begin{cases} -\epsilon_{AS} \tilde{s}_{yy} \rho^{-2}, & \tilde{s}_{yy} > 0, \\ 0, & \tilde{s}_{yy} \leq 0. \end{cases} \quad (15)$$

The tensor is rotated back to yield (R_{xx}, R_{xy}, R_{yy}) in the global frame.

3.2.3 Injection into stress divergence

A kernel-ratio weight is computed:

$$f_{ab} = \left(\frac{W_{ij}}{W_{\Delta p}} \right)^n, \quad n \in \mathbb{N}, \quad W_{\Delta p} > 0. \quad (16)$$

Then the pairwise artificial stress is

$$\mathbf{R}_{ij} = f_{ab} (\mathbf{R}_i + \mathbf{R}_j). \quad (17)$$

A representative component form of the discrete divergence consistent with the implementation is:

$$\begin{aligned} (\nabla \cdot \boldsymbol{\sigma})_{x,i} \approx & \sum_{j \in \mathcal{N}(i)} m_j \left(\frac{\sigma_{xx,i}}{\rho_i^2} + \frac{\sigma_{xx,j}}{\rho_j^2} + R_{xx,ij} \right) \frac{\partial W_{ij}}{\partial x} \\ & + \sum_{j \in \mathcal{N}(i)} m_j \left(\frac{\sigma_{xy,i}}{\rho_i^2} + \frac{\sigma_{xy,j}}{\rho_j^2} + R_{xy,ij} \right) \frac{\partial W_{ij}}{\partial y}. \end{aligned} \quad (18)$$

3.3 Damping / dissipation mechanisms

3.3.1 Artificial viscosity (Monaghan-type)

Define relative kinematics

$$\mathbf{v}_{ij} = \mathbf{v}_i - \mathbf{v}_j, \quad \mathbf{x}_{ij} = \mathbf{x}_i - \mathbf{x}_j. \quad (19)$$

Viscosity activates only under approach:

$$\mathbf{v}_{ij} \cdot \mathbf{x}_{ij} < 0. \quad (20)$$

Define averaged quantities

$$h_{ij} = \frac{1}{2}(h_i + h_j), \quad \rho_{ij} = \frac{1}{2}(\rho_i + \rho_j), \quad c_{ij} = \frac{1}{2}(c_i + c_j), \quad (21)$$

with wave speed (as used in the code)

$$c_i = \sqrt{\frac{K(T_i)}{\rho_i}}. \quad (22)$$

Then

$$\mu_{ij} = \frac{h_{ij} (\mathbf{v}_{ij} \cdot \mathbf{x}_{ij})}{\|\mathbf{x}_{ij}\|^2 + \eta_{AV}^2 h_{ij}^2}, \quad (23)$$

$$\Pi_{ij} = \frac{-\alpha_{AV} c_{ij} \mu_{ij} + \beta_{AV} \mu_{ij}^2}{\rho_{ij}}. \quad (24)$$

The acceleration contribution is

$$\dot{\mathbf{v}}_i += - \sum_{j \in \mathcal{N}(i)} m_j \Pi_{ij} \nabla W_{ij}. \quad (25)$$

3.3.2 Tool contact penalty (normal) and friction (tangential)

Let g_N be penetration depth and $\mathbf{n}$ the surface normal returned by the tool contact query. With particle mass m_s and timestep Δt :

$$\mathbf{f}_N = -\alpha_C m_s \frac{g_N}{\Delta t^2} \mathbf{n}. \quad (26)$$

Friction uses a clamped update with coefficient μ :

$$\|\mathbf{f}_T\| \leq \mu \|\mathbf{f}_N\|. \quad (27)$$

The resulting forces enter the momentum equation as external forces:

$$\dot{\mathbf{v}}_i += \frac{\mathbf{f}_{c,i} + \mathbf{f}_{t,i}}{m_i}. \quad (28)$$

3.3.3 XSPH kinematic smoothing (stabilization)

XSPH modifies the position RHS with parameter ϵ_{XSPH} and $\rho_{ij} = \frac{1}{2}(\rho_i + \rho_j)$:

$$\dot{\mathbf{x}}_i += -\epsilon_{XSPH} \sum_{j \in \mathcal{N}(i)} \frac{m_i}{\rho_{ij}} (\mathbf{v}_i - \mathbf{v}_j) W_{ij}. \quad (29)$$

4 Results

At each timestep, after assembling RHS terms at the half-step state, the solver produces:

- Pressure field p_i via (9).
- Artificial stress correction $(R_{xx,i}, R_{xy,i}, R_{yy,i})$ via (11)–(15) and coupling (16)–(17).
- Internal acceleration contributions from stress divergence (e.g., (18)) and artificial viscosity (25).
- External contact/friction forces and their contribution to $\dot{\mathbf{v}}_i$ via (28).
- Updated state u^{n+1} via (7).

4.1 Boundary conditions

Fixed particles are enforced by resetting to reference position (X_i, Y_i) and zeroing velocities and kinematic RHS terms:

$$\mathbf{x}_i \leftarrow \mathbf{X}_i, \quad \mathbf{v}_i \leftarrow \mathbf{0}, \quad \dot{\mathbf{x}}_i \leftarrow \mathbf{0}, \quad \dot{\mathbf{v}}_i \leftarrow \mathbf{0}. \quad (30)$$

5 Conclusions

- Pressure is computed by a temperature-dependent linear EOS (9) and enters the momentum balance through $\boldsymbol{\sigma} = \mathbf{S} - p\mathbf{I}$ (10) inside a symmetric SPH stress divergence.
- Artificial stress is a tensile-instability stabilization: it detects tensile principal-like stresses via rotation (11)–(13), applies a repulsive correction only in tension (14)–(15), and injects it into the stress divergence with a kernel-ratio weighting (16)–(17).
- Damping/dissipation is dominated by Monaghan artificial viscosity active only for approaching particles (20)–(25), and penalty contact plus friction forces from the tool (26)–(28). XSPH provides additional stabilization by smoothing advection velocities (29).
- Coupling is explicit and operator-split, evaluated at the half step, and advanced by leapfrog (6)–(7).